# Algorithm Design for Resilient Cyber-Physical Systems using an Automated Attack Generative Model

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***Abstract*—This paper presents a suite of algorithms for detecting and localizing attacks in cyber-physical systems, and for performing resilient state estimation with a Kalman filter monitored by a chi-square detector. High performance of the underlying detection and localization algorithms is achieved by generating training data that cover a large region of the attack space. An unsupervised generative model trained by physics-based discriminators—constructed from the Kalman-filter innovation and closed-loop critical-output maps—is designed to generate successful false data injection attacks. The generated adversarial examples train a multi-class deep network that detects and localizes compromised measurements. A pruning algorithm then improves localization precision and provides probabilistic guarantees linking attack sparsity, post-pruning observability, Kalman-filter estimation error, and detection precision. The framework is validated on a water distribution cyber-physical system.**

***Index Terms*—Resilient estimation, Kalman filter, chi-square detector, false data injection attack, generative model, multilayer perceptron, pruning, observability**

## I. Introduction

Modern cyber-physical critical infrastructures (CPCI) are facing fast-evolving cyber threats. Cyberattacks can strongly impact the operation of the CPCIs. For instance, a suspected cyber intruder took control of the Prykarpattyaoblenergo power system control center in western Ukraine in December 2015, leaving 230,000 people without electricity for up to 6 hours. That was the first time hackers have successfully targeted a country's power grid [**?**]. Also, cyber attacks in water systems have already become a reality. In 2015, 25 cyber attacks were disclosed in several water systems [**?**]. And recently in 2020, a malicious cyber-attack attempted to raise the chlorine level in Israel's water supply to a dangerous proportion [**?**]. Supervisory Control and Data Acquisition (SCADA) is a critical part of the CPCI that is highly susceptible to cyberattacks. It is responsible for collecting measurements from Remote Terminal Units (RTU) and sending them to the Control Centers (CC) to perform various functions such as contingency analysis, optimal planning, state estimation (SE), etc. The primary purpose of those functions is to maintain stable and secure operation of the CPCI.

SE is a core function in the CCs which monitors the system states using information collected from the SCADA systems and other measurement devices. SE can also detect and identify bad data that can cause significant error in the resulting estimate. The SE algorithm's accuracy depends on how pure and accurate the system measurements are. If any malicious or erroneous measurements pass through undetected, it could mislead the operators into making catastrophically wrong decisions. Numerous studies have shown that traditional SE algorithms are vulnerable to False Data Injection Attacks (FDIAs), where an intruder aims to hack multiple RTUs or even communication channels to insert fake measurements in order to misguide the CCs operational process [**?**], [**?**], [**?**], [**?**]. Since FDIA was introduced in [**?**], different researchers have proposed various techniques for detection and localization, including diagnostic robust generalized potential [**?**], generalized likelihood ratio [**?**], fast Go-Decomposition (GoDec) [**?**], Markov chain [**?**], Bayesian detection with binary hypothesis [**?**], modified unbiased finite impulse response (UFIR) estimator [**?**], and cosine similarity matching scheme [**?**] and some residual-based approaches such as the Kullback Leibler distance method [**?**], unscented Kalman filter [**?**], $\chi^2$ failure detector [**?**] and a residual-based localization scheme [**?**]. However, due to their dependence on the system model, the associated model uncertainty would affect their accuracy. Thus, if the FDIAs are designed appropriately, these model-based or residual-based detectors can easily be bypassed [**?**], [**?**]. Take for example, filter-based attack detector designs [**?**], [**?**]; the recovered attack signals could have deviations due to the uncertainties of the underlying model used in the filter design. Thus, if the attacker substitutes current meter measurements with historical readings following the same distribution, the Bayesian framework and generalized likelihood ratio approaches are unable to identify this FDIA [**?**].

To overcome the limitations of traditional residual-based

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bad data detection approaches, data-driven solutions based on machine learning algorithms have been widely adopted for detecting and localizing FDIAs due to their fast execution times and accurate results [?], [?], [?], [?], [?], [?], [?]. [?] utilized and tested various machine learning algorithms in detecting FDIAs. The results showed that machine learning algorithms can detect FDIAs accurately and faster. [?] proposed a deep neural network algorithm that can automatically detect and localize FDIAs in power systems. The proposed method can integrate the underlying graph topology of the grid and spatial correlations of its measurement data to jointly detect and localize the FDIAs in power systems. [?] proposed a hierarchy of neural networks with the capability of detecting novel attacks by training networks to understand the entire normal space instead of attack space.

In addition to attack detection and localization, resilient estimation approaches focus on maintaining the correctness of state estimates in the presence of adversarial attacks, which enables resilient operation of the CPSs [?]. Although attacks are possibly unbounded and can be designed to "fool" $\ell_2$ observers [?], [?], the attack signals are sparse due to the resource limitation of the attackers. Based on these characteristics, the authors in [?] proposed an $\ell_1$ minimization program for estimator design on linear systems and proposed an upper-bound on the total number of compromised measurements for guaranteed secure estimation. Some extension work considered robustness on resilient $\ell_1$ estimation in presence of modeling error [?], worst-case estimation error bound in presence of bounded noise [?], and $\ell_1 - \ell_2$ estimation scheme [?]. And some iterative resilient observer designs has been proposed for determinant system, such as event-trigger Luenburger Observer [?] and Gramian-based estimator [?]. For stochastic system, several attack-resilient control strategies have also been studied [?], [?]. In addition, resilient estimation and control design has also been studied for nonlinear systems [?].

Furthermore, some enhanced resilient CPS designs have been studied to accommodate even worse adversarial condition. To push the limit on the number of compromised sensors below which secure estimation would be guaranteed, additional prior information could be used in the resilient estimation design. Existing results have used such information as measurement prior, support prior, and state prior: In [?], three forms of *state prior* are considered: *sparsity information*, $(\alpha, \overline{n}_0)$ *sparsity information,*, and *side information.* These are information generally unavailable to the malicious agent. In [?], the authors designed an improved resilient estimator, with input and state constraints, against actuator attacks. In our previous work [?], [?], [?], the resiliency of $\ell_1$ based observers is enhanced by utilizing a measurement prior, generated by a pre-trained Gaussian process. In [?], a resilient unscented Kalman filter was proposed based on improved support prior. In [?], a weighted $\ell_1$ observer is designed to exploit the support prior more rigorously.

Alternatively, improved resiliency of CPS could also be achieved by using more accurate attack detection and localization algorithms. For data-driven attack detectors, especially supervised learning-based detectors, exploiting more of the attack space and system nominal conditions could improve accuracy significantly [?]. This has motivated researchers to study attack data generation strategies for improving the detection/localization boundaries of the underlying data-driven algorithms. The goal of the attack generation problem is to generate feasible attacks that could elicit significant targeted disruption on system, while bypassing bad data detection. Model-based attack generation strategies often assume the attackers have the knowledge of system models [?], [?]. This assumption limits the application range of those attack generation strategies. Thus, automated attack generation without knowledge of the system models, is currently gaining a lot of attention. Although this topic has been studied widely in the machine learning area, for example; adversarial machine learning [?] and deepfake [?], the automated attack generation problem for CPS is a new challenging problem since the physical process dynamics is included. Inspired by the generative adversarial network (GAN) [?], some earlier researches [?], [?], [?] have studied GAN-based attack generation by using an existing attack dataset. The idea is to use GAN to learn the underlying distribution of feasible attacks, then a richer dataset of attacks could be generated. However, it is not practical to assume the existence of a non-generative attack dataset. Thus, an automated attack generation process, without attack examples, is needed. To the best of the authors knowledge, this paper is the first paper to study this problem.

*Contribution*: This paper proposes an enhanced resilient solution for CPSs subject to FDIAs. It is a hybrid physics-based and data-driven, detection-based resilient estimation framework built around a *Kalman filter* with a *chi-square* innovation detector. The framework consists of an automated attack generator, supervised learning-based attack detection and localization, support pruning, and a resilient KF on the retained safe sensors. To train the supervised detector, abundant adversarial examples are required. Rather than relying on expensive optimization-based generators [?], [?] or labeled attack corpora [?], [?], we propose an unsupervised generative model trained by physics-based discriminators derived from the KF innovation (stealthiness against $\chi^2$) and the closed-loop critical-output map (effectiveness). Secondly, a pruning algorithm systematically improves localization precision, with guarantees quantified as a function of the pruning reliability level. Thirdly, we establish a *quantitative bridge* among attack sparsity, post-pruning observability, KF mean-square estimation error (via the subset Riccati covariance), and detection/localization precision ($\mathsf{PPV}_\eta$ and the $\chi^2$ false-alarm specification). Consequently, the CPS can maintain certified estimation performance while attacks are underway.

The remainder of the paper is organized as follows. Section ?? collects notation. Section ?? introduces the stochastic CPS model, the Kalman filter, and the chi-square detector. Section ?? defines successful FDIAs against this pair and presents the physics-guided generative model. Section ?? develops MLP-based detection/localization, pruning, and the sparsity–observability–precision bridge. Section ?? designs the resilient KF and links subset covariances to pruning reliability. Section ?? reports water-distribution simulations, and Section ?? concludes.

## II. Notation and Preliminary

The following notations and definitions are used throughout the paper: $\mathbb{R}, \mathbb{R}^n, \mathbb{R}^{n\times m}$ denote the space of real numbers, real vectors of length $n$, and real matrices of $n$ rows and $m$ columns respectively. $\mathbb{R}_+$ denotes the space of positive real numbers. Normal-face lower-case letters ($e.g.\ x \in \mathbb{R}$) are used to represent real scalars, bold-face lower-case letters ($e.g.\ \mathbf{x} \in \mathbb{R}^n$) represent vectors, while normal-face upper-case letters ($e.g.\ X \in \mathbb{R}^{n\times m}$) represent matrices. $X^\top$ denotes the transpose of the matrix $X$. The spectral radius of the square matrix $X$ is denoted by $\rho(X)$. $X^\dagger \triangleq \left(X^\top X\right)^{-1} X^\top$ denotes the Moore-Penrose inverse. $\mathbf{1}_n$ and $I_n$ denote the vector of ones and the identity matrix of size $n$ respectively. Let $\mathcal{T} \subseteq \{1,\dots,n\}$, then, for a matrix $X \in \mathbb{R}^{m\times n}$, $X_\mathcal{T} \in \mathbb{R}^{|\mathcal{T}|\times m}$ is the sub-matrix obtained by extracting the rows of $X$ corresponding to the indices in $\mathcal{T}$. $\mathcal{T}^c$ denotes the complement of a set $\mathcal{T}$, and the universal set on which it is defined will be clear from the context. We use $I_T$ (or simply $I$ when $T$ is clear from context) to represent a $T$-time window $[i-T+1, i]$. In the same vein, $I-1$ is used to denote the time window $[i-T, i-1]$ accordingly. The support of a vector $\mathbf{x} \in \mathbb{R}^n$ is the set of the indices of nonzero entries in $\mathbf{x}$, defined as $\mathsf{supp}(\mathbf{x}) \triangleq \{i \subseteq \{1,\dots,n\} | \mathbf{x}_i \neq 0\}$. A vector $\mathbf{x} \in \mathbf{R}^n$ is said to be $k$-sparse if $|\mathsf{supp}(\mathbf{x})| \leq k$, and $\Sigma_k$ denotes the subset of $k$-sparse vectors. A moving-horizon vector $\mathbf{x}_I \in \Sigma_k$ is given by $\mathbf{x}_I = \begin{bmatrix}\mathbf{x}_1^\top & \cdots & \mathbf{x}_T^\top\end{bmatrix} \in \mathbb{R}^{nT}$, where each constituent vector $\mathbf{x}_j \in \Sigma_k, j \in I$. $\mathsf{argsort}\downarrow(\mathbf{x})$ denotes a function that returns the sorted indices of vector $\mathbf{x}$ in descending order of the magnitude of $\mathbf{x}_i$. The symbol $\odot$ denotes element-wise multiplication of two vectors, as in $\mathbf{z} = \mathbf{x} \odot \mathbf{y}$, where $\mathbf{z}_i = \mathbf{x}_i\mathbf{y}_i$. The operator $\|\mathbf{z}\|_{1,\mathbf{w}} \triangleq \sum_{i=1}^n \mathbf{w}_i\mathbf{z}_i$ is a weighted 1-norm of a vector $\mathbf{z} \in \mathbb{R}^n$, with the weight vector $\mathbf{w} \in \mathbb{R}^n$. A continuous function $\phi : [0, a) \to [0, \infty)$ is said to belong to class $\mathcal{K}$ if it is strictly increasing and $\phi(0) = 0$. Both $e^x$ and $\exp(x)$ are used to represent the exponential function. The symbol $*$ denotes the convolution operator of two vectors; given two vectors $\mathbf{u} \in \mathbb{R}^m$ and $\mathbf{v} \in \mathbb{R}^n$,

$$\mathbf{u} * \mathbf{v} \triangleq \sum_j \mathbf{u}_j \mathbf{v}_{k+1-j}.$$

The symbol $\mathbb{E}$ denotes the expectation operator of a random variable. $z \sim \mathcal{B}(1,p)$ represents a Bernoulli distributed random variable $z$, with $\mathbb{E}[z] = p$. Thus, the sum of independent Bernoulli random variables $\{z_1, z_2, \cdots, z_N\}$ with $z_i \sim \mathcal{B}(1,p_i)$ satisfies the Poisson-binomial distribution, and the closed-form expression of the probability density function (pdf) of $\sum_i z_i$ is given by [**?**]:

$$\mathsf{Pr}\left\{\sum_{i=1}^N z_i = k\right\} = \mathbf{r}_k, \quad k = 0, \cdots, N, \tag{1}$$

where

$$\mathbf{r} = \prod_{i=1}^N p_i \cdot \begin{bmatrix}-s_1\\1\end{bmatrix} * \begin{bmatrix}-s_2\\1\end{bmatrix} * \cdots * \begin{bmatrix}-s_N\\1\end{bmatrix}, \tag{2}$$

with $s_i = -\frac{1-p_i}{p_i}$. The following lemma presents a result that will be useful for subsequent developments in this paper.

**Lemma II.1.** *For any non-decreasing convex function $\Phi : \mathbb{R} \to \mathbb{R}_+$ with $\Phi(0) = 1$,*

$$\mathbb{E}[\Phi(z)] \geq \mathsf{Pr}\{z > 0\}.$$

*Proof.* Based on the definition of expectation,

$$\begin{aligned}\mathbb{E}[\Phi(z)] &= \sum_{x\in\mathbb{R}} \Phi(x)\mathsf{Pr}\{z = x\}\\ &= \sum_{x\leq 0} \Phi(x)\mathsf{Pr}\{z = x\} + \sum_{x>0} \Phi(x)\mathsf{Pr}\{z = x\}.\end{aligned}$$

Since $\Phi(x) > 0$, then

$$\mathbb{E}[\Phi(z)] \geq \sum_{x>0} \Phi(x)\mathsf{Pr}\{z = x\}.$$

Also, since $\Phi$ is non-decreasing and $\Phi(0) = 1$, we have that $\Phi(x) \geqslant 1 \ \ \forall x > 0$. Thus,

$$\mathbb{E}[\Phi(z)] \geq \sum_{x>0} \mathsf{Pr}\{z = x\} = \mathsf{Pr}\{z > 0\}.$$

□

## III. Model Development

The following stochastic linear model is used to describe the physical process of a CPS:

$$\begin{aligned}\mathbf{x}_{i+1} &= A\mathbf{x}_i + B\mathbf{u}_i + \mathbf{w}_i,\\ \mathbf{y}_i &= C\mathbf{x}_i + \mathbf{v}_i,\end{aligned} \tag{3}$$

where $\mathbf{x}_i \in \mathbb{R}^n$ is the state vector, $\mathbf{u}_i \in \mathbb{R}^p$ the control input, and $\mathbf{y}_i \in \mathbb{R}^m$ the sensor measurement. The process noise $\mathbf{w}_i \sim \mathcal{N}(\mathbf{0}, Q)$ and measurement noise $\mathbf{v}_i \sim \mathcal{N}(\mathbf{0}, R)$ are mutually independent white sequences with $Q \succeq 0$ and $R \succ 0$. The appropriately dimensioned system matrices $A, B$ and $C$ are assumed to satisfy the following:

- A1. The pair $(A, B)$ is controllable.
- A2. There exists a positive integer $k_0 \leq m$ such that the pair $(A, C_\mathcal{S})$ is observable for all $\mathcal{S} \subset \{1, 2, \cdots, m\}$ with $|\mathcal{S}| \geq k_0$.
- A3. The pair $(A, Q^{1/2})$ is stabilizable.

Assumption A2 encodes a quantitative redundancy level of the sensing network: any subset of at least $k_0$ sensors preserves observability. This sparsity–observability threshold will be linked to estimation error and detection precision in Sections **??** and **??**.

### A. Kalman filter

Under A2–A3, the minimum-variance linear estimator of (**??**) is the Kalman filter (KF). Let $\hat{\mathbf{x}}_{i|i-1}$ and $P_{i|i-1}$ denote the predicted state and covariance. The prediction and update steps are

$$\hat{\mathbf{x}}_{i|i-1} = A\hat{\mathbf{x}}_{i-1|i-1} + B\mathbf{u}_{i-1}, \tag{4}$$
$$P_{i|i-1} = AP_{i-1|i-1}A^\top + Q, \tag{5}$$
$$\mathbf{r}_i = \mathbf{y}_i - C\hat{\mathbf{x}}_{i|i-1}, \tag{6}$$
$$S_i = CP_{i|i-1}C^\top + R, \tag{7}$$
$$K_i = P_{i|i-1}C^\top S_i^{-1}, \tag{8}$$
$$\hat{\mathbf{x}}_{i|i} = \hat{\mathbf{x}}_{i|i-1} + K_i\mathbf{r}_i, \tag{9}$$
$$P_{i|i} = (I - K_iC)\, P_{i|i-1}. \tag{10}$$

The innovation $\mathbf{r}_i$ in (**??**) is zero-mean Gaussian with covariance $S_i$ in the attack-free case. When the filter operates in steady state, $P_{i|i-1} \to P^-$, $K_i \to K$, and $S_i \to S = CP^-C^\top + R$, where $P^-$ solves the discrete algebraic Riccati equation associated with $(A, C, Q, R)$. For notational convenience we write $\hat{\mathbf{x}}_i \triangleq \hat{\mathbf{x}}_{i|i}$ for the filtered estimate used by the controller.

### B. Chi-square bad-data detector

Because $\mathbf{r}_i \sim \mathcal{N}(\mathbf{0}, S_i)$ under the nominal hypothesis $\mathcal{H}_0$ (no attack), the normalized innovation squared

$$g_i \triangleq \mathbf{r}_i^\top S_i^{-1} \mathbf{r}_i \tag{11}$$

follows a chi-square distribution with $m$ degrees of freedom. A chi-square detector with false-alarm rate $\beta$ is

$$\mathsf{BDD}(\mathbf{y}_i) = \begin{cases} 0, & \text{if } g_i \le \tau_{\chi^2}(\beta), \\ 1, & \text{otherwise,} \end{cases} \tag{12}$$

where the threshold $\tau_{\chi^2}(\beta)$ is the $(1-\beta)$-quantile of $\chi^2_m$. Equivalently, over a moving window $I = [i-T+1, i]$ one may use the cumulative statistic $g_I = \sum_{j \in I} g_j \sim \chi^2_{Tm}$ under $\mathcal{H}_0$. Unlike a purely residual-norm test, (**??**) accounts for the innovation covariance and admits an explicit probabilistic false-alarm specification.

### C. Closed-loop control

The following certainty-equivalence control law is considered:

$$\mathbf{u}_i = K_p \hat{\mathbf{x}}_i, \tag{13}$$

where the gain $K_p$ is designed so that $\rho(A + BK_p) \in (0, 1)$. Substituting (**??**) into (**??**) yields

$$\mathbf{x}_{i+1} = (A + BK_p)\,\mathbf{x}_i + BK_p\,(\hat{\mathbf{x}}_i - \mathbf{x}_i) + \mathbf{w}_i.$$

Hence, when the KF estimation error remains mean-square bounded, the closed-loop state is mean-square bounded as well. In particular, under A1–A3 and attack-free measurements, $\mathbb{E}\|\hat{\mathbf{x}}_i - \mathbf{x}_i\|_2^2 = \mathrm{tr}(P_{i|i})$ converges to the steady-state filtered covariance, and the origin of the mean closed-loop dynamics is asymptotically stable [**?**], [**?**].

## IV. Automated Attack Generation

In this section, a generative model is developed for producing successful FDIAs against the KF and chi-square detector of Section **??**. Including an additive attack $\mathbf{e}_i \in \mathbb{R}^m$ in the measurement equation yields the adversarial model

$$\mathbf{y}_i = C\mathbf{x}_i + \mathbf{v}_i + \mathbf{e}_i. \tag{14}$$

Under attack, the innovation becomes

$$\mathbf{r}_i = C\tilde{\mathbf{x}}_{i|i-1} + \mathbf{v}_i + \mathbf{e}_i, \tag{15}$$

where $\tilde{\mathbf{x}}_{i|i-1} = \mathbf{x}_i - \hat{\mathbf{x}}_{i|i-1}$ is the prediction error. Consequently, the filtered estimation error is driven by

$$\tilde{\mathbf{x}}_{i|i} = (I - K_iC)\tilde{\mathbf{x}}_{i|i-1} - K_i\mathbf{v}_i - K_i\mathbf{e}_i. \tag{16}$$

Over a moving window $I = [i-T+1, i]$, stack the attacks as $\mathbf{e}_I = \begin{bmatrix}\mathbf{e}_{i-T+1}^\top & \cdots & \mathbf{e}_i^\top\end{bmatrix}^\top$. Linearizing about the steady-state KF $(K, S)$ and collecting the affine maps from $\mathbf{e}_I$ to the windowed normalized innovation and to the induced critical-output bias defines the physics-based discriminators used below.

### A. Chi-square stealthiness and critical-output effectiveness

Attackers aim to disturb a critical functional output $\mathbf{y}_c = C_c\mathbf{x}$ without triggering the chi-square detector (**??**). The windowed chi-square statistic induced by $\mathbf{e}_I$ (neglecting the zero-mean noise contribution in the discriminator design) is characterized by the quadratic form

$$\|M_2\mathbf{e}_I\|_2^2 \triangleq \sum_{j \in I} \mathbf{e}_j^\top S^{-1}\mathbf{e}_j = \mathbf{e}_I^\top \big(\mathrm{blkdiag}(S^{-1}, \ldots, S^{-1})\big)\mathbf{e}_I, \tag{17}$$

so that stealthiness requires $\|M_2\mathbf{e}_I\|_2$ to remain below a design threshold tied to $\tau_{\chi^2}$. (When a cumulative detector $g_I$ is used, the same map applies with the corresponding block-diagonal weighting.)

For effectiveness, substitute the control law (**??**) and the attack-driven KF error (**??**) into the closed-loop dynamics. With $\bar{A} = A + BK_p$ and $\bar{B} = BK_p$, the critical output satisfies, to first order in the attack,

$$\mathbf{y}_{c,i+1} = C_c\bar{A}\mathbf{x}_i + C_c\bar{B}\,M_K\mathbf{e}_I + \text{noise terms}, \tag{18}$$

where $M_K$ is the linear map from the windowed attack to the filtered estimation error induced by the steady-state KF gain (explicitly, $M_K$ is built by unrolling (**??**) over $I$). Thus the attack effectiveness is measured by $\|M_1\mathbf{e}_I\|_2$ with $M_1 = C_c\bar{B}M_K$. From the attacker's perspective, $\|M_1\mathbf{e}_I\|_2$ should be large while $\|M_2\mathbf{e}_I\|_2$ stays small.

**Definition 1** ($(\alpha,\epsilon)$-Successful FDIA)**.** *Consider the CPS in* (**??**) *with the KF* (**??**)*–*(**??**) *and chi-square detector* (**??**)*. Given a positive integer $k < m$, the attack sequence $\mathbf{e}_I \in \Sigma_k \subset \mathbb{R}^{Tm}$ is said to be $(\alpha,\epsilon)$-successful if*

$$\|M_1\mathbf{e}_I\|_2 \ge \alpha \quad \textit{and} \quad \|M_2\mathbf{e}_I\|_2 \le \epsilon \tag{19}$$

*for some $\alpha, \epsilon \in \mathbb{R}_+$.*

**Remark 1.** *The first criterion requires a minimum disruption of the critical output through the KF–controller loop. The second requires the attack-induced contribution to the (windowed) chi-square statistic to remain below $\epsilon$, thereby encoding stealthiness against* (**??**)*. Feasibility-based definitions of this type are preferred over pure maximizations of effectiveness under a hard stealth constraint [?], [?].*

Here $k$ is the attack sparsity per time index. Classical model-based FDIAs that fool KF/$\chi^2$ monitors [**?**] or stronger $\ell_1$ estimators [**?**] are often computationally heavy for online data generation. We therefore use a generative model, in the spirit of GANs [**?**], to sample from the feasible set

$$\mathcal{S}(\alpha,\epsilon) \triangleq \big\{\mathbf{e} \;\big|\; \|M_1\mathbf{e}\|_2 \ge \alpha,\ \|M_2\mathbf{e}\|_2 \le \epsilon\big\}. \tag{20}$$

**Definition 2.** *Given a set $\mathcal{S} \subset \mathbb{R}^n$ and a prior $P_z(\mathbf{z})$, a generative model is a mapping $f : \mathbf{z} \mapsto \mathbb{R}^n$ such that $f(\mathbf{z}) \in \mathcal{S}$ with high probability.*

We take $G(\mathbf{z};\boldsymbol{\theta}_g)$ as a neural network with parameters $\boldsymbol{\theta}_g$, driven by $\mathbf{z} \sim P_z$. Physics-based discriminators evaluate $\|M_1\mathbf{e}\|_2$ and $\|M_2\mathbf{e}\|_2$ on the generator output, as in Fig. **??**.

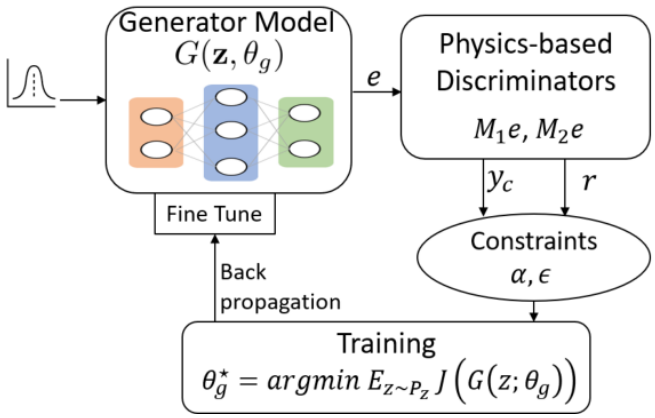



Fig. 1. Physics-based generative model and training process ($\alpha$ is the lower-bound constraint for attack effectiveness against the KF–controller loop, while $\epsilon$ is the upper-bound constraint for stealthiness against the chi-square detector)

The indicator-like training loss is

$$J(\mathbf{e}) = \mathsf{ReLU}(\alpha - \|M_1\mathbf{e}\|_2) + \mathsf{ReLU}(\|M_2\mathbf{e}\|_2 - \epsilon), \quad (21)$$

and the generator is trained by

$$\boldsymbol{\theta}_g^* = \arg\min \mathop{\mathbb{E}}_{\mathbf{z}\sim\mathbf{P}_z} J\big(G(\mathbf{z};\boldsymbol{\theta}_g)\big) \quad (22)$$

via back-propagation. The resulting samples populate the adversarial training set used in Section **??**.

## V. Detection and Localization

Through an automated attack generation by a well-trained generative model, an adversarial training dataset could be generated and used to improve the precision of the supervised learning-based attack detection and localization algorithm. In this section, a detection and localization algorithm is given by training a multi-layer perceptron (MLP) on the generated attack dataset. In addition, by analyzing the uncertainty of the MLP's output, a pruning algorithm is proposed to improve the localization precision further.

The term `attack location` refers to sensor channels (or indices) through which the attack signals are added to the system. Let $\mathcal{T}$ denote the actual attack support, then we define its indicator element-wise as:

$$\mathbf{q}_i \begin{cases} 0 & \text{if } i \in \mathcal{T} \\ 1 & \text{otherwise.} \end{cases} \quad (23)$$

Next, the definitions of detection and localization are given:

**Definition 3** (Detection)**.** *Given a measurement vector $\mathbf{y} \in \mathbb{R}^m$, detection is a function $D : \mathbb{R}^m \to \{0,1\}$ whose output $D(\mathbf{y})$ indicates the existence of attack signal in the input $\mathbf{y}$.*

**Definition 4** (Localization)**.** *Given a measurement vector $\mathbf{y} \in \mathbb{R}^m$, localization is a function $L : \mathbb{R}^m \to \{0,1\}^m$ whose output indicates detection result for each measurement channel. In particular, it is an estimate $\hat{\mathcal{T}}$ of the actual attack support $\mathcal{T}$. Its indicator $\hat{\mathbf{q}}$ is defined similarly to (**??**), using $\hat{\mathcal{T}}$*

As universal function approximators, shown in Cybenko's theorem [**?**], MLPs are used to describe mathematical models via a regression scheme. Attack detection and localization is a multi-line binary classification task. With adequate training dataset generated by the automated attack generator, MLP's advantages of being a supervised learning network could be fully utilized.

The MLP used in this paper is a three-layer neural network containing two *LeakyReLU* hidden layers and one *Sigmoid* output layer, and the final classification outputs are 0 (for attack) if the corresponding MLP output is less than 0.5, and 1 (for no attack) otherwise. The output layer's size is the same as the input size and equal to the number of attacks. Consequently, the MLP is given by

$$g(\mathbf{y};\boldsymbol{\theta}) \triangleq \phi\left(\sum_{i=1}^{n} w_i\mathbf{y}_i + b\right), \quad (24)$$

where $\phi$ is the activation function of the output layer and $\boldsymbol{\theta}$ is a vector of tunable weights and biases. Then, the localization output is given by

$$\hat{\mathbf{q}}_i = \begin{cases} 1 & \text{if } g(\mathbf{y}_i;\boldsymbol{\theta}) \geq 0.5 \\ 0 & \text{otherwise.} \end{cases} \quad (25)$$

Next, a supervised learning algorithm, via back propagation, is used to train the MLP using the binary cross-entropy loss function:

$$J(\boldsymbol{\theta}) = \frac{1}{N}\sum_{i=1}^{N}\Big(\mathbf{q}^{(i)}\log g(\mathbf{y}^{(i)};\boldsymbol{\theta}) + (\mathbf{1}-\mathbf{q}^{(i)};\boldsymbol{\theta})\log(\mathbf{1}-g(\mathbf{y}^{(i)};\boldsymbol{\theta}))\Big), \quad (26)$$

where $(\mathbf{y}^{(i)}, \mathbf{q}^{(i)})$ is the $i$th training data, and $N$ is the total number of data points.

Since the MLP-based attack detection and localization method described above is a multi-line binary classification, the agreement between the localization result $\hat{\mathbf{q}}$ and the actual attack support $\mathbf{q}$ can be described by a Bernoulli distributed agreement variable $\epsilon_i \sim \mathcal{B}(1,\mathbf{p}_i)$ given by

$$\mathbf{q}_i = \epsilon_i\hat{\mathbf{q}}_i + (1-\epsilon_i)(1-\hat{\mathbf{q}}_i), \quad (27)$$

where $\mathbf{p}_i = \mathbb{E}[\epsilon_i] = \mathsf{Pr}\{\epsilon_i = 1\}$ is the confidence of the localization algorithm at $i$ channel, which is often obtained by the true rate in *Receiver Operator Characteristic* (ROC) of the algorithm.

Consequently, the precision of the resulting support prior indicator $\hat{\mathbf{q}}$ is defined as follows:

**Definition 5** (Positive Prediction Value, Precision, PPV [**?**])**.** *Given an estimate $\hat{\mathbf{q}} \in \{0,1\}^N$ of an unknown attack support indicator $\mathbf{q} \in \{0,1\}^N$, PPV is the proportion of $\mathbf{q}$ that is correctly identified in $\hat{\mathbf{q}}$. It is given by*

$$PPV = \frac{\|\mathbf{q}\odot\hat{\mathbf{q}}\|_0}{\|\hat{\mathbf{q}}\|_0}. \quad (28)$$

Next, with the attack localization result $\hat{\mathcal{T}}$, the estimated support of safe nodes can be represented as $\hat{\mathcal{T}}^c =$

$\{j_1, j_2, \cdots, j_{|\hat{\mathcal{T}}^c|}\}$. A *Pruning Algorithm* is proposed to prune the most probable erroneous localization results in $\hat{\mathcal{T}}^c$.

The first step is to obtain a trust length $l_\eta (\leq Tm)$ based on the confidence of the localization results $\mathbf{p}$. Given a reliability level $\eta \in (0,1)$, the trust length $l_\eta$ is an integer defined as the maximum number of safe nodes correctly localized in $\hat{\mathcal{T}}^c$ with a probability of at least $\eta$;

$$l_\eta \triangleq \max\left\{k \middle| \Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i \geq k\right\} \geq \eta\right\}. \tag{29}$$

Let

$$\bar{\mathbf{r}} \triangleq \begin{bmatrix} r_0 \\ r_1 \\ \vdots \\ r_{|\hat{\mathcal{T}}^c|} \end{bmatrix} = \prod_{i=1}^{|\hat{\mathcal{T}}^c|} \mathbf{p}(j_i) \begin{bmatrix} -s_{j_1} \\ 1 \end{bmatrix} * \begin{bmatrix} -s_{j_2} \\ 1 \end{bmatrix} * \cdots * \begin{bmatrix} -s_{j_{|\hat{\mathcal{T}}^c|}} \\ 1 \end{bmatrix}, \tag{30}$$

where $s_{j_i} = -\frac{1-\mathbf{p}_i}{\mathbf{p}_i}$. Then, $\Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i = k\right\} = \bar{\mathbf{r}}_k$. Thus (**??**) becomes[1]

$$\begin{aligned} l_\eta &= \max\left\{k \middle| \Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i \leq k\right\} \leq 1-\eta\right\} \\ &= \max\left\{k \middle| \sum_{i=0}^{k} \bar{\mathbf{r}}_i \leq 1-\eta\right\}. \end{aligned} \tag{31}$$

The second step is to obtain a *Pruned Support Prior* $\hat{\mathcal{T}}^c_\eta$ of size $l_\eta$. This is done through the robust extraction:

$$\hat{\mathcal{T}}^c_\eta = \left\{\text{argsort}\downarrow(\mathbf{p} \odot g(\mathbf{y};\boldsymbol{\theta}))\right\}_1^{l_\eta}, \tag{32}$$

where $\{\cdot\}_1^{l_\eta}$ denotes an index extraction from the first elements to $l_\eta$ elements. To obtain a pruned support prior consistent with the model assumptions in Section **??**, its size should satisfy $l_\eta \geq k_0$, where $k_0$ is a lower bound of $\hat{\mathcal{T}}^c_\eta$ such that $(A, C_{\hat{\mathcal{T}}^c_\eta})$ is observable. This can be guaranteed by the proper choice of the reliability level $\eta$, as given in the following theorem.

**Theorem V.1.** *Given a measurement redundancy preservation parameter $k_0 > 0$, the size of pruned support prior satisfies $l_\eta \geq k_0$ if*

$$\eta \leq 1 - \sum_{i=0}^{|\hat{\mathcal{T}}^c|} e^{k_0 - i} \bar{\mathbf{r}}_i, \tag{33}$$

*where $\bar{\mathbf{r}}$ is given by (**??**), and $\hat{\mathcal{T}}^c$ is the estimated support of safe nodes from the attack localization algorithm.*

[1] $\Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i \leq k\right\} \leq 1-\eta$ is equivalent to $\Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i = 0\right\} + \Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i = 1\right\} + \cdots \Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i = k\right\} \leq 1-\eta$ since $k$ could only be integer.

*Proof.* According to (**??**), a sufficient condition for $l_\eta \geq k_0$ is

$$\Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i \geq k_0\right\} \geq \eta,$$

which is equivalent to:

$$\Pr\left\{k_0 - \sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i \leq 0\right\} \geq \eta$$

and

$$\Pr\left\{k_0 - \sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i \geq 0\right\} \leq 1-\eta. \tag{34}$$

According to Lemma **??**, let $z = k_0 - \sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i$, and take $\Phi(z) = \exp(z)$ yields a sufficient condition for (**??**) as

$$\begin{aligned} &\mathbb{E}\left[\exp\left(k_0 - \sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i\right)\right] \leq 1-\eta, \\ \equiv\ & e^{k_0}\mathbb{E}\left[\exp\left(-\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i\right)\right] \leq 1-\eta. \end{aligned} \tag{35}$$

Since $\Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i = k\right\} = \bar{\mathbf{r}}_k$, it follows that

$$\begin{aligned} \mathbb{E}\left[\exp\left(-\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i\right)\right] &= \sum_{k=0}^{|\hat{\mathcal{T}}^c|} e^{-k} \Pr\left\{\sum_{i\in\hat{\mathcal{T}}^c} \epsilon_i = k\right\} \\ &= \sum_{k=0}^{|\hat{\mathcal{T}}^c|} e^{-k} \bar{\mathbf{r}}_k. \end{aligned} \tag{36}$$

Combining (**??**) and (**??**) yields

$$\sum_{i=0}^{|\hat{\mathcal{T}}^c|} e^{k_0 - i} \bar{\mathbf{r}}_i \leq 1-\eta. \tag{37}$$

Finally, an upper bound on $\eta$ is obtained as (**??**). □

**Remark 2.** *From (**??**), it is clear that, necessarily, the upper bound should be strictly positive. This yields the following pruning requirement on the localization performance parameters:*

$$\sum_{i=0}^{|\hat{\mathcal{T}}^c|} e^{-i} \bar{\mathbf{r}}_i < e^{-k_0}. \tag{38}$$

*If this holds, according to assumption A2, with at least $k_0$ safe nodes remaining in the pruned support prior $\hat{\mathcal{T}}^c_\eta$, $(A, C_{\hat{\mathcal{T}}^c_\eta})$ is observable. Consequently, a Kalman filter designed on that subset is well-posed, and the mean-square estimation error remains bounded as formalized in Section **??**.*

Moreover, it was shown in [**?**] that the precision $\text{PPV}_\eta$ of the pruned support prior $\hat{\mathcal{T}}^c_\eta$ given by (**??**) achieves

$$\Pr\{\text{PPV}_\eta = 1\} \geq \eta. \tag{39}$$

This indicates that, with probability at least $\eta$, every sensor in $\hat{\mathcal{T}}_\eta^c$ is safe. Combined with Remark **??**, $(A, C_{\hat{\mathcal{T}}_\eta^c})$ is then observable, which is the sensing condition required by the resilient Kalman filter in Section **??**.

### A. Quantitative bridge: sparsity, observability, and precision

We now make explicit the chain connecting attack sparsity, post-pruning observability, localization precision, and—in the next section—KF estimation error. Let $k = |\mathcal{T}|$ be the number of compromised sensors at a time index, and write $m_s = |\hat{\mathcal{T}}^c|$ for the number of sensors declared safe by the localization stage before pruning.

**Proposition 1** (Sparsity–observability budget)**.** *Suppose at most $k$ sensors are attacked and the localization stage returns a safe-set estimate of size $m_s$. If the pruning length satisfies $l_\eta \geq k_0$, then a necessary sensing budget is*

$$m_s \geq l_\eta \geq k_0 \quad \Rightarrow \quad k \leq m - k_0 + \big(m_s - (m-k)\big)^+, \tag{40}$$

*where $(\cdot)^+ = \max\{\cdot, 0\}$ accounts for false-safe declarations. In particular, when localization makes no false-safe error on attacked channels, observability after pruning requires $k \leq m - k_0$.*

*Proof.* At most $m - k$ sensors are truly safe. The pruned set has size $l_\eta$ and is contained in the declared safe set of size $m_s$. Enforcing $l_\eta \geq k_0$ for A2 yields $m_s \geq k_0$. If every attacked sensor is correctly excluded, then $m_s \leq m - k$, hence $k \leq m - k_0$. False-safe inclusions relax this bound by the displayed surplus term. □

**Proposition 2** (Precision–observability trade-off)**.** *Fix localization confidences $\{\mathbf{p}_i\}$ and $k_0$. Let $\eta^\star$ be the right-hand side of* (**??**)*. Then every reliability level $\eta \in (0, \eta^\star]$ simultaneously guarantees*

(i) *$l_\eta \geq k_0$ (post-pruning observability), and*
(ii) *$\Pr\{\mathsf{PPV}_\eta = 1\} \geq \eta$ (clean pruned sensors).*

*Increasing $\eta$ toward $\eta^\star$ raises the guaranteed precision but shortens $l_\eta$; decreasing $\eta$ lengthens the pruned set (and typically reduces the KF error covariance derived in Section* **??***) at the expense of a weaker precision certificate.*

*Proof.* Claim (i) is Theorem **??**; claim (ii) is (**??**). The monotonicity of $l_\eta$ in $\eta$ follows from (**??**) because the admissible $k$ in the $\max$ decreases as $\eta$ increases. □

**Remark 3** (Chi-square detection vs. localization precision)**.** *The chi-square test* (**??**) *provides a* global *detection decision with false-alarm rate $\beta$, whereas $\mathsf{PPV}_\eta$ quantifies* channel-wise *localization precision after pruning. Stealthy $(\alpha, \epsilon)$-FDIAs are generated to keep the attack contribution to $g_i$ below $\epsilon$; the MLP+pruning pipeline then recovers a high-precision safe set on which the KF operates. The two notions of "detection precision" are therefore complementary: $\chi^2$ bounds miss/false-alarm at the innovation level, while $\mathsf{PPV}_\eta$ bounds contamination of the estimator's sensor subset.*

Finally, Algorithm **??** summarizes the attack detection and localization pipeline with automated attack generation and pruning.

**Algorithm 1:** Attack detection and localization with automated attack generation and pruning

I. **Offline: Dataset generation and training.**
1) Given time horizon $T$, prepare physics-based discriminator maps $M_1$ (KF–controller effectiveness) and $M_2$ (chi-square stealthiness) ← (**??**), (**??**).
2) Given the number of attacks $|\mathcal{T}|$, randomize attack support $\mathcal{T}$, and obtain its indicator $\mathbf{q}$ ← (**??**).
3) Given attack indexes $\alpha, \epsilon$, train the generative model (GM) by solving the optimization problem in (**??**).
4) Generate attack dataset $\{Y\}$ using the trained GM. Evaluate generated attack dataset via simulation.
5) Return the effective subset of attack dataset as the training dataset $\{Y, \mathbf{q}\}$.
6) Train the MLP in (**??**) using the generated dataset $\{Y, \mathbf{q}\}$ via backpropagation.

II. **Online: Localization and pruning**
1) Input real-time measurements $\mathbf{y}$, obtain the classification result $\hat{\mathbf{q}}$ ← (**??**), (**??**);
2) Obtain the pruned support prior $\hat{\mathcal{T}}_\eta^c(\hat{\mathbf{q}}_\eta)$ ← (**??**), (**??**).

**Output**: $\hat{\mathcal{T}}_\eta^c$ (pruned support prior)

## VI. Resilient Estimator

In this section, the Kalman filter of Section **??** is redesigned on the pruned safe support produced by Algorithm **??**, in order to maintain secure state estimation under FDIAs. The scheme is shown in Fig. **??**: attack detection/localization returns an estimate $\hat{\mathcal{T}}^c$ of the safe sensors; pruning downselects the most trustworthy subset $\hat{\mathcal{T}}_\eta^c$; the resilient KF then updates using only $\mathbf{y}_{\hat{\mathcal{T}}_\eta^c}$, with the corresponding output matrix $C_{\hat{\mathcal{T}}_\eta^c}$ and noise covariance $R_{\hat{\mathcal{T}}_\eta^c}$.

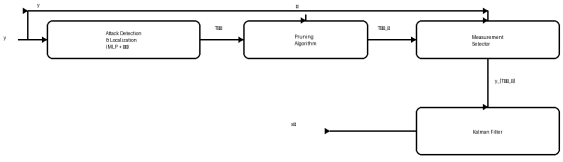



Fig. 2. Schematic diagram of the resilient Kalman filter with chi-square monitoring and support pruning

### A. Resilient Kalman filter

Given $\hat{\mathcal{T}}_\eta^c$ at time $i$, the resilient KF uses the reduced measurement model

$$\mathbf{y}_{\hat{\mathcal{T}}_\eta^c, i} = C_{\hat{\mathcal{T}}_\eta^c}\mathbf{x}_i + \mathbf{v}_{\hat{\mathcal{T}}_\eta^c, i} + \mathbf{e}_{\hat{\mathcal{T}}_\eta^c, i}, \tag{41}$$

and applies (**??**)–(**??**) with $(C, R)$ replaced by $(C_{\hat{\mathcal{T}}_\eta^c}, R_{\hat{\mathcal{T}}_\eta^c})$. Denote the resulting filtered estimate and covariance by $\hat{\mathbf{x}}_i'$ and $P_{i|i}(\hat{\mathcal{T}}_\eta^c)$. By (**??**), $\mathbf{e}_{\hat{\mathcal{T}}_\eta^c, i} = \mathbf{0}$ with probability at least $\eta$.

When $|S| \geq k_0$ and $(A, C_S)$ is observable, the subset KF admits a unique stabilizing steady-state predicted covariance

$P^-(S) \succ 0$ solving the DARE for $(A, C_S, Q, R_S)$, with filtered covariance

$$P_\infty(S) = (I - K(S)C_S)P^-(S), \quad (42)$$
$$K(S) = P^-(S)C_S^\top \left(C_S P^-(S) C_S^\top + R_S\right)^{-1}.$$

The map $S \mapsto \operatorname{tr} P_\infty(S)$ is monotonically nonincreasing under set inclusion: if $S_1 \subset S_2$ and both satisfy A2, then $\operatorname{tr} P_\infty(S_2) \le \operatorname{tr} P_\infty(S_1)$ [**?**]. This yields the graded estimation-error link in the sparsity–observability–precision chain.

**Theorem VI.1** (Resilient KF performance)**.** *Let $\hat{\mathcal{T}}_\eta^c$ be given by* (**??**)*, and suppose $\eta$ satisfies* (**??**) *with the localization performance criterion* (**??**)*. Then, with probability at least $\eta$,*

(i) $\mathbf{e}_{\hat{\mathcal{T}}_\eta^c} = \mathbf{0}$ *and* $|\hat{\mathcal{T}}_\eta^c| \ge k_0$*, so* $(A, C_{\hat{\mathcal{T}}_\eta^c})$ *is observable;*

(ii) *the resilient KF coincides with the attack-free KF on the same sensor subset, and its steady-state mean-square error satisfies*

$$\lim_{i\to\infty} \mathbb{E}\left\|\hat{\mathbf{x}}_i' - \mathbf{x}_i\right\|_2^2 = \operatorname{tr} P_\infty(\hat{\mathcal{T}}_\eta^c); \quad (43)$$

(iii) *relative to the full-sensor attack-free KF,*

$$\operatorname{tr} P_\infty(\{1,\dots,m\}) \le \operatorname{tr} P_\infty(\hat{\mathcal{T}}_\eta^c) \le \operatorname{tr} P_\infty(S_{k_0}), \quad (44)$$

*for every observable subset $S_{k_0}$ of size $k_0$ that can arise from pruning at reliability $\eta$.*

*Proof.* Claim (i) follows from (**??**), Theorem **??**, and assumption A2. On that event the reduced measurements equal the attack-free subset measurements, so the KF recursions—and hence the limiting covariance (**??**)—coincide with the attack-free subset filter, which is claim (ii). Claim (iii) is the standard information monotonicity of the Riccati map under additional measurements, together with the worst-case size-$k_0$ subset admitted by the pruning length constraint. □

**Corollary VI.1.1** (Bridge summary)**.** *For attack sparsity $k$, observability threshold $k_0$, and localization confidences $\{\mathbf{p}_i\}$, Propositions* **??**–**??** *and Theorem* **??** *jointly imply: every $\eta \in (0, \eta^\star]$ yields a pruned set that is clean and observable with probability $\ge \eta$, and the KF error is then exactly $\operatorname{tr} P_\infty(\hat{\mathcal{T}}_\eta^c)$, sandwiched between the full-sensor and worst $k_0$-sensor Riccati costs. Thus sparsity constrains feasible $(k, k_0)$, precision $\eta$ trades against set size (hence against $\operatorname{tr} P_\infty$), and estimation error is quantitatively tied to both.*

The chi-square monitor (**??**) may still be evaluated on the innovation of the resilient KF (with $S_i$ computed from $C_{\hat{\mathcal{T}}_\eta^c}$) to provide an online global alarm, while the MLP+pruning path supplies the channel support required by (**??**)–(**??**).

## VII. Simulation

In this section, the resiliency improvement due to automated attack generation is shown in simulation by implementing Algorithm **??** and the resilient Kalman filter of Theorem **??** on a water distribution system. The experiments also illustrate the sparsity–observability–precision–error bridge of Corollary **??**.

The water distribution system, shown in Fig. **??**, contains 10 distributed operating tanks and 1 storage tank. The goal

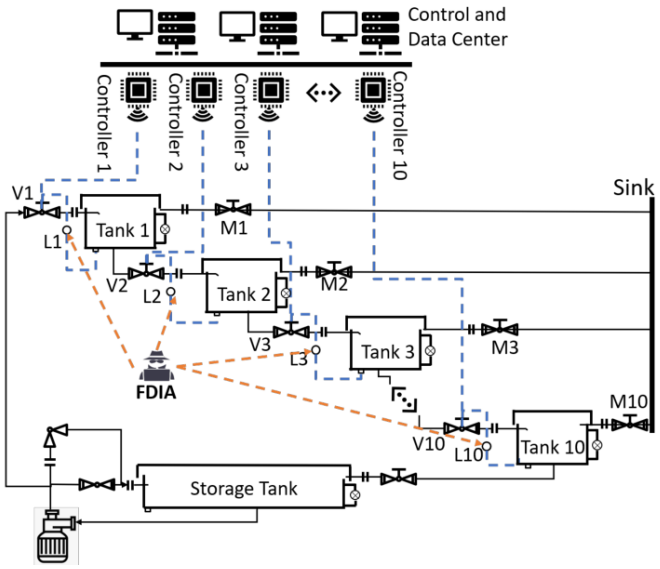



Fig. 3. Block diagram depiction of a water distribution system under FDIAs (black solid are water pipelines, blue dotted lines are wireless data transmission lines for sensors data and control commands, orange dotted lines are attack injection route)

is to regulate all operating tanks' water levels around desired values. The magnetic valves $V_1 - V_{10}$ at the inlet of operating tanks are controlled to adjust the water level of water tanks, and the manual valves $M_1 - M_{10}$ can be manipulated, at the demand point, to use the water from the corresponding tanks. The effect of these valves is modelled as the random demands $\mathbf{d}$ in (**??**). The valves at the storage tank are fixed at a constant opening value. For each operating tank, two sensors-water level meter and pressure sensors ($L_1$ $L_{10}$), are installed. The pressure sensors measure the difference of water level between adjoining tanks. The water levels adjustment process is approximated by the LTI model:

$$\mathbf{x}_{i+1} = A\mathbf{x}_i + B\mathbf{u}_i - \mathbf{d}_i,$$
$$\mathbf{y}_i = C\mathbf{x}_i + \mathbf{v}_i + \mathbf{e}_i. \quad (45)$$

where, $\mathbf{x}, \mathbf{u} \in \mathbb{R}^{10}$, $\mathbf{y}, \mathbf{v} \in \mathbb{R}^{61}$ are vector of water level, control input of magnetic valves, sensor measurements, and measurement noise respectively, $\mathbf{d} \in \mathbb{R}^{10}$ is the vector of random demands, and $\mathbf{e} \in \mathbb{R}^{61}$ is the injected FDIAs. For the sake of clean presentation, interested readers are directed to https://github.com/ZYblend/Resilient_WDS_AutoAttackGen for the values of system parameters $A \in \mathbb{R}^{10\times10}, B \in \mathbb{R}^{10\times10}, C \in \mathbb{R}^{61\times10}$. Given a desired water level $\mathbf{x}_d = 0.5 * \mathbf{1}_{10}$, the proportional controller below is used;

$$\mathbf{u}_i = -K(\hat{\mathbf{x}} - \mathbf{x}_d) - B^{-1}(A - I_{10})\mathbf{x}_d.$$

The critical measurement in this scenario is the sum of all water levels of operating tanks, thus, the critical measurement matrix is given by $C_c = [1\ 1\ \cdots\ 1] \in \mathbb{R}^{1\times10}$. Firstly, a simulation experiment is done to assess the observability of the simulated system in order to obtain the least number of measurements from which the system states are fully observable. A Monte Carlos experiment, with 50000 samples of randomly selected measurement subsets $\mathcal{S}$, is performed to calculated the

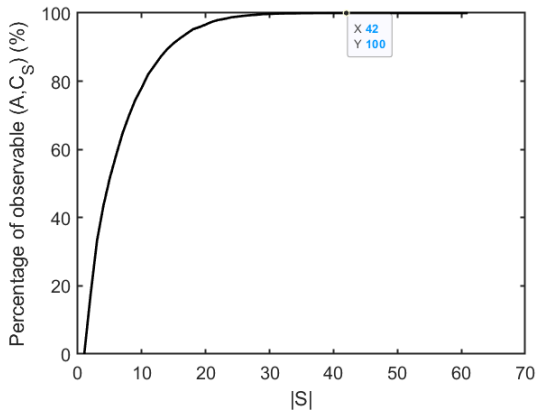



Fig. 4. Observability of simulated systems. The percentage of observable $(A, C_{\mathcal{S}})$ is calculated as the ratio of the number of observable $(A, C_{\mathcal{S}})$ to the total number of samples (=50000).

probability of full observerability of $(A, C_{\mathcal{S}})$ using $|\mathcal{S}|$. Fig **??** shows that full observerability is preserved with $k_0 \geq 42$, corresponding to at most 30% pruned measurements.

Next, we execute automated attack generation (against the KF/$\chi^2$ pair), MLP training, and Monte-Carlo simulation of the resilient KF under different detection/localization strategies.

### A. Automated attack generation

A fixed time step of $T_s = 0.01s$ is used and the moving-window size is set as $T = 8T_s$. With different numbers of attacks, the discriminator maps $M_1$ (critical-output effectiveness through the KF–controller loop) and $M_2$ (chi-square stealthiness) are assembled for each random attack support. The generator in (**??**) is then trained as in Fig. **??**. Due to measurement noise and random demands, the thresholds are chosen as $\alpha = 0.1$ and $\epsilon = 0.5$ (with $\epsilon$ aligned to the design chi-square level). Fig. **??** shows effectiveness and the stealthiness statistic of generated attacks during training relative to $(\alpha, \epsilon)$. The trained generator maps uniform inputs to feasible attacks in (**??**).

The trained generator is then used to generate a dataset $D_{\mathcal{S}}(\alpha, \epsilon)$ of attacks. In order to inject the generated attacks, recursive feasibility [**?**] should be considered. An online moving-horizon attack injection scheme used here is to solve the following sample-based optimization program at each time step $i$:

$$\begin{aligned} \underset{\mathbf{e}_i}{\text{Maximize}} \quad & \left\| M_1 \begin{bmatrix} \mathbf{e}^- \\ \mathbf{e}_i \end{bmatrix} \right\|_2 \\ \text{Subject to:} \quad & \left\| M_2 \begin{bmatrix} \mathbf{e}^- \\ \mathbf{e}_i \end{bmatrix} \right\|_2 \leq \epsilon \\ & \mathbf{e}_i \in D_{\mathcal{S}}(\alpha, \epsilon) \end{aligned} \tag{46}$$

where $\mathbf{e}^-$ is the attack history during the time window $I^- = [i - T + 1, i - 1]$. Fig. **??** shows the performance of the generated attacks under the injection program (**??**). The left panel shows that the windowed chi-square / innovation statistic under attack stays close to the nominal (attack-free) trajectory, i.e., the attacks remain stealthy relative to (**??**). The right panel shows a clear deviation of the critical measurement, confirming effectiveness through the KF–controller loop.

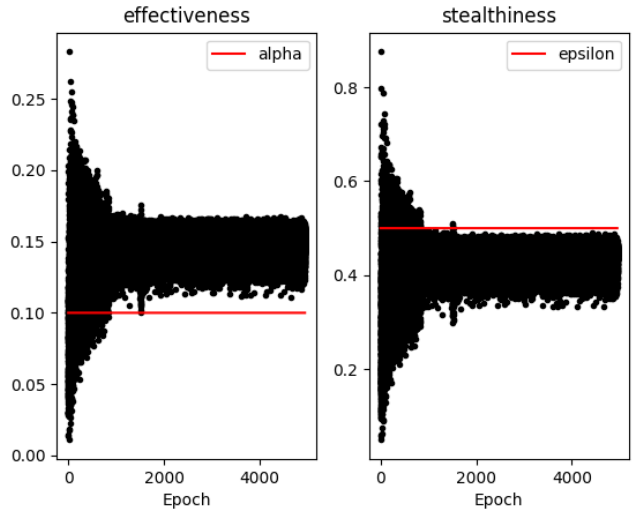



Fig. 5. Feasibility of generated attacks during training. $\epsilon$ upper-bounds the chi-square stealthiness statistic $\|M_2 e\|_2$; $\alpha$ lower-bounds KF–controller effectiveness $\|M_1 e\|_2$.

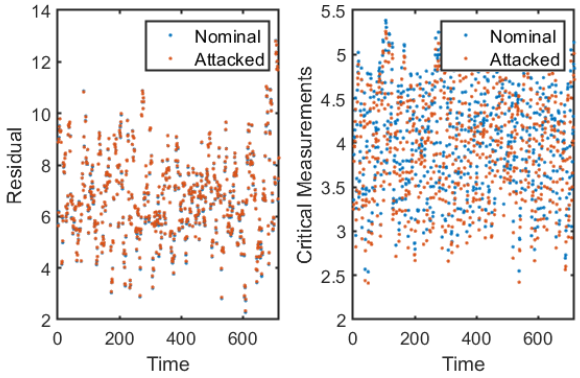



Fig. 6. Performance of generated attacks injected through 39% measurement nodes. (Left: windowed innovation / chi-square stealthiness statistic; Right: critical measurement.)

Next, the above automated attack generation scheme is trained to generate different types of attacks including different attack supports and different number of attacks. The generated attacks are then used to train the supervised learning-based attack detection and localization algorithm in next subsection.

### B. MLP training

To show one of the merits of the proposed automated attack generation scheme, two training datasets are prepared: one produced by random additive signals on measurements and the other augments the dataset with attack samples generated by the trained generator. Next, two MLP classifiers are trained on two datasets respectively. MLP1 and MLP2 are both four-layer fully-connected neural network with three "relu" hidden layers of $512, 1024, 512$ neurons respectively. The output layer has 61 neurons and uses "sigmoid" activation function. MLP1 is trained by the first random attack dataset, while MLP2 is trained the second training dataset containing automated generated attacks. The testing F1 score and precision of MLP1 are 85.1% and 99.7% respectively, while MLP2 achieves 88.7%

and 99.1% respectively. The testing datasets used is 10% of the corresponding datasets, so the testing datasets for MLP2 and MLP1 are different. Next, we compare the localization precision of the two MLPs on a same testing dataset which is an equal mixture of dataset 1 and dataset 2. As shown in Fig. **??**, MLP2 has successfully shifted the localization precision to the right, indicating a better performance overall. However, MLP1's performance is nearly uniform for different localization precision higher than 0.5. This is because it is trained by random attack dataset. The mean precision of MLP1 is 74.37%, and the mean precision of MLP2 is 98.69% on a testing set of successful FDIAs.

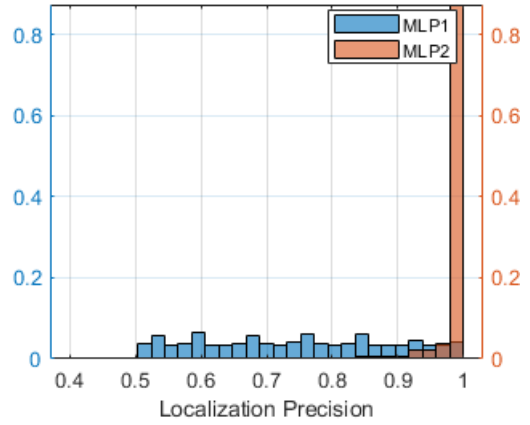



Fig. 7. A sample-based comparison of the performance of two MLP classifiers on same testing set of successful FDIAs. (MLP1: trained only with random attack generation, MLP2: trained with random attack generation and the proposed automated attack generation, 50000 test cases are used.)

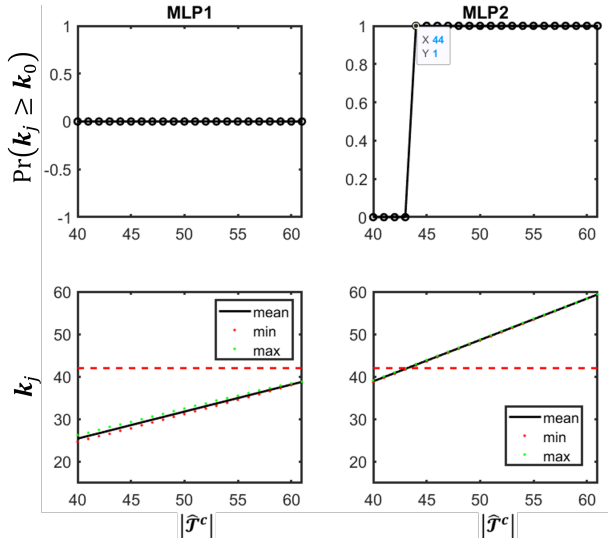



Fig. 8. A sample-based comparison of post-pruning observability for MLP1 and MLP2. (50000 test cases are used. The y axis for the top two subplots $Pr(k_j \geq k_0)$ is the sample estimate of the probability of at least $k_0$ measurements remaining after pruning. The black lines in the bottom subplots show the mean value of $k_j$ that is the number of measurements remaining after pruning, while the red and green dots are the minimum and maximum values of $k_j$'s respectively.

As shown in Fig. **??**, full observability can be guaranteed with certainty if $k_0 \geq 42$ measurements remains after pruning. For this, we evaluate the sufficient condition in (**??**) on MLP1 and MLP2. First, the two MLPs are run on 10000 cases generated by the trained generative model. From the results, the confidence vectors of the MLPs are estimated by calculating the true rates at all 61 measurement nodes; for node $i$, $\mathbf{p}_i = \frac{TP_i+TN_i}{P_i+N_i}$. Next, a sample based experiment is performed. For each support size $|\hat{\mathcal{T}}_j^c| \in \{40, 41, \ldots, 61\}$, we generated $\min\left\{{}^{61}C_{|\hat{\mathcal{T}}_j^c|}, 50000\right\}$ unique supports $\hat{\mathcal{T}}^c$, and evaluated the right hand side of (**??**) for each unique support as follows:

$$k_j = -\ln\left(\sum_{i=0}^{|\hat{\mathcal{T}}_j^c|} e^{-i}\bar{\mathbf{r}}_i\right),$$

where the Poisson-binomial probability density vector $\bar{\mathbf{r}}_i$ is calculated, using the confidence vectors of the corresponding MLP, based on (**??**). For each $|\hat{\mathcal{T}}_j^c| \in \{40, 41, \ldots, 61\}$, and each MLP, the sample probability of the event $k_j \geq k_0$ and the expected value of $k_j$ are shown in Fig. **??**. The results show that MLP2 has a bigger chance of preserving post-pruning observability. Indeed, according to the figure, post-pruning observability cannot be guaranteed when MLP1 is used. In addition, from the bottom plots in Fig. **??**, it is seen that MLP2 reduces the uncertainty on the post-pruning observability compared to MLP1. By considering the sample-based analysis in Fig. **??** and Fig. **??** together, it is seen that MLP performs better when trained with data generated by the proposed automated attack generation.

Next, we compare MLP1 and MLP2 inside the resilient KF scheme.

### C. Resilient Kalman filter

The trained MLPs, with and without pruning, are embedded in the real-time WDS simulation with the resilient KF of Section **??**. A Monte-Carlo experiment estimates: (1) the **probability of sustaining normal operation** (pSNO)—the fraction of times the critical-measurement deviation stays below a threshold; (2) the **probability of triggering an alarm** (pTA)—the fraction of times the chi-square / innovation statistic exceeds its attack-free baseline by a threshold. Larger pSNO and pTA indicate better resiliency.

Attacks with sparsity from 1 to 30 are considered (five random-support scenarios each). For each detector variant (MLP1, MLP1+pruning, MLP2, MLP2+pruning), the pipeline in Fig. **??** is simulated. Let $g_I$ denote the windowed chi-square statistic and $\mathbf{y}_{c_{i+1}}$ the critical output. Normalized deviations

$$\Delta g = \frac{|(g_I - g_d) - (g_I^\star - g_d)|}{\max\left(|(g_I - g_d) - (g_I^\star - g_d)|\right)},$$

$$\Delta \mathbf{y} = \frac{\left|(\mathbf{y}_{c_{i+1}} - \mathbf{y}_{c_d}) - (\mathbf{y}^\star_{c_{i+1}} - \mathbf{y}_{c_d})\right|}{\max\left(\left|(\mathbf{y}_{c_{i+1}} - \mathbf{y}_{c_d}) - (\mathbf{y}^\star_{c_{i+1}} - \mathbf{y}_{c_d})\right|\right)},$$

are formed relative to the attack-free noise-free baselines $g_d$ and $\mathbf{y}_{c_d} = C_c\mathbf{x}_d$. Then $\text{pTA} = \frac{1}{N}\|\Delta g \geq \tau_1\|_{\ell_0}$ and $\text{pSNO} = \frac{1}{N}\|\Delta \mathbf{y} \leq \tau_2\|_{\ell_0}$, with $\tau_1 = 1\%$ and $\tau_2 = 3\%$ calibrated on attack-free runs. These metrics empirically probe Corollary **??**: as sparsity $k$ grows, preserving $|\hat{\mathcal{T}}_\eta^c| \geq k_0 = 42$ and high

$\mathsf{PPV}_\eta$ becomes harder, which appears as degraded pSNO/pTA unless localization (MLP2) is strong enough.

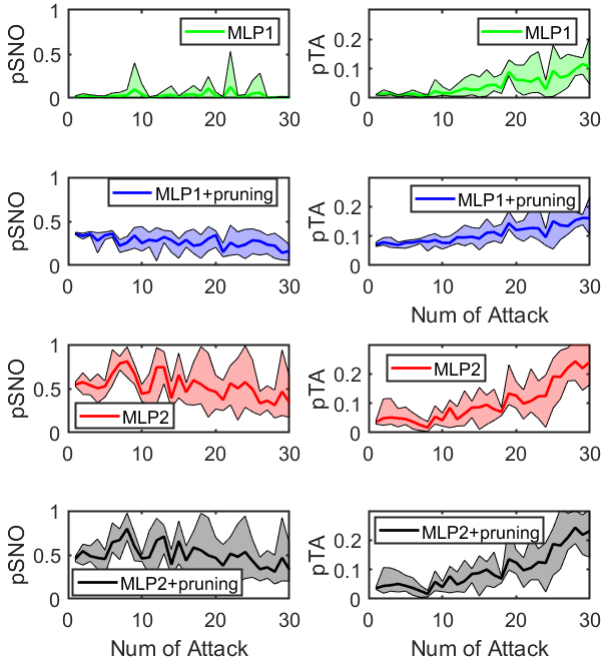



Fig. 9. Resilient KF performance under different attack detection strategies; larger pSNO and pTA indicate better resiliency. Solid lines are means; shaded regions indicate spread.

Fig. **??** compares pSNO and pTA across strategies. MLP2 outperforms MLP1 (and even MLP1+pruning), showing the value of generator-augmented training. Mean localization precision (**??**) versus number of attacks is shown in Fig. **??**. Pruning still raises MLP2 precision, but state-estimation gains saturate once the observability budget $k_0$ binds—consistent with the $\eta$–$l_\eta$–$\mathrm{tr}\, P_\infty$ trade-off in Proposition **??** and Theorem **??**. MLP2+pruning nevertheless tightens the spread of pSNO/pTA, i.e., reduces performance uncertainty.

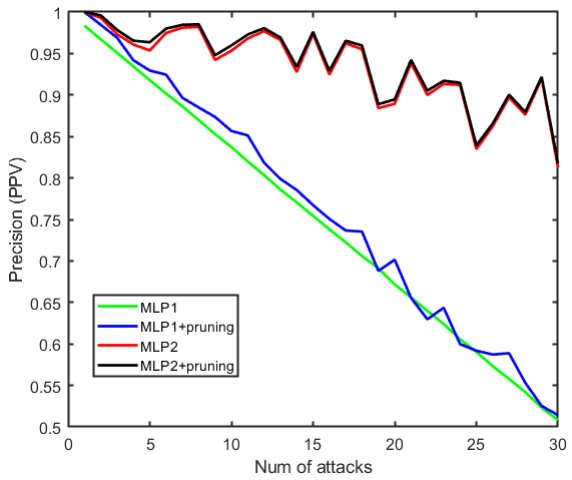



Fig. 10. A comparison of the mean precision of different attack detection strategies in the system simulation versus different number of attacks

Finally, Fig. **??** shows a time-domain comparison under the same 39% compromised measurements as in Fig. **??**. Schemes using MLP2 (with or without pruning) reduce critical-output deviation (smaller $\Delta \mathbf{y}$) while increasing the chi-square / innovation excursion (larger $\Delta g$), i.e., attacks become both less effective and less stealthy once the resilient KF is fed a high-precision safe set.

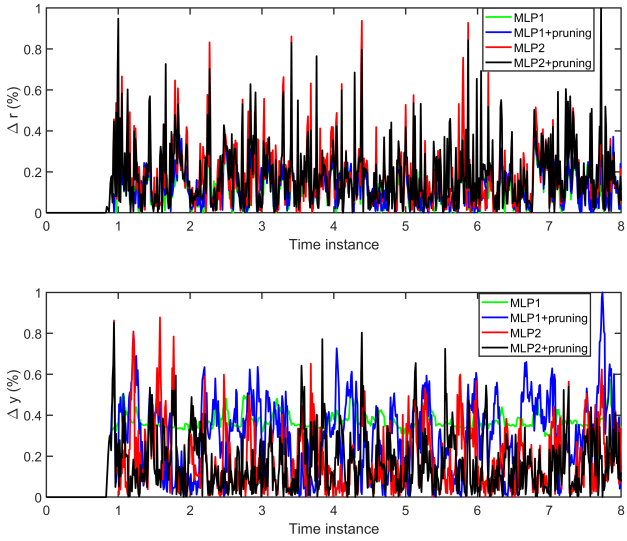



Fig. 11. Time-domain resilient KF results for four localization strategies under the attack scenario of Fig. **??**.

## VIII. Conclusion

In this paper, we presented a resilient estimation framework for cyber-physical systems under false data injection attacks, centered on a Kalman filter with a chi-square innovation detector. Resiliency is improved by training the attack detector with samples from an automated, physics-guided generative model whose discriminators encode KF–controller effectiveness and $\chi^2$ stealthiness—without requiring prior labeled attack corpora. An MLP localizes compromised channels; a pruning stage raises localization precision with an explicit reliability parameter $\eta$. We further established a quantitative bridge among attack sparsity, the observability threshold $k_0$, the KF subset error $\mathrm{tr}\, P_\infty(S)$, and detection precision ($\mathsf{PPV}_\eta$ and the $\chi^2$ false-alarm rate). Simulations on a water distribution CPS support the theory: generator-augmented training improves localization and closed-loop resiliency, while pruning reduces performance variability subject to the observability budget.

Open directions include analyzing generator convergence under uncertain physics discriminators, learning data-driven discriminators with low online cost, extending the sparsity–error–precision bridge to nonlinear filters, and integrating weighted robust estimators in the loop with the same pruning certificates.